

§1

THE SITUATION

For a decade, finding a drone meant finding its radio, and an entire industry was built on that assumption. The fibre-optic drone ended it. A glass thread carries its commands and video, so it transmits nothing at all — nothing to detect, nothing to jam, nothing to trace back. Its airframe is fibreglass and returns almost nothing to radar, and in our heat it is the same temperature as the air around it. These are not prototypes: they were used across Ukraine through 2025 and 2026, trialled by the US Marine Corps in January 2026, and reported in April 2026 as being flown specifically to beat radar. One of them can fly over Singapore today and no system we own would know it was there.

PROBLEM STATEMENT

Ideate creative ways to detect something that emits no radio, returns almost nothing to radar, and sits at the temperature of the air around it. Showcase a demo that reports how often it cries wolf against a real background — not only how far it can see.

§2

THINGS TO TAKE NOTE OF

01

NON-RF DETECTION MODALITIES

Sound, the light it blocks or reflects, the disturbance it makes in somebody else's radio waves, and the cable it trails. Take one further than it has been taken.

02

FALSE ALARM RATE

That figure decides whether an operator leaves the system switched on. A detector raising two false alerts an hour is disconnected regardless of range, and here the sky is full of legitimate drones. DSTA's published work with Thales is directed at precisely this.

03

ACOUSTIC DETECTION IN HIGH AMBIENT NOISE

Acoustic detection is the most promising answer to a target that transmits nothing, but published ranges assume quiet. Beneath the Changi approach, beside an expressway or next to the port they do not hold. Acoustic detection in genuinely loud environments is an open research question, not a solved technique.

04

PASSIVE SENSING AND SENSOR SURVIVABILITY

Passive sensing removes that exposure but yields bearing, not range. Two separated sensors are then required, plus a method of deciding their detections are the same object. That association problem is substantial and is almost never set.

05

COVERAGE ECONOMICS

Everyone asks for a better sensor. Almost nobody asks how an area is covered — siting, power, backhaul, unit cost. A cheap sensor affordable in thousands defeats a good one affordable in threes.

06

CUED VERSUS UNCUED SEARCH

Searching a directed sector and watching an entire sky differ by orders of magnitude. Most work assumes a cue and collapses without one. State which you are addressing.

§3

TWO EXAMPLES

ONE — CHANGE THE MODALITY, COMPETE ON COST

Two engineers put a microphone and a mobile telephone on a pole. That became a national network of thousands of nodes at a few hundred dollars each, contributing a meaningful share of all detections — not because acoustic sensing is good, but because it is cheap enough to place everywhere. The move was refusing to compete on sensor quality at all.

TWO — PURSUE WHAT IS ADJACENT TO THE TARGET

One approach to the fibre-optic drone does not attempt to detect the drone. It flies a camera along logistics routes hunting the cable, and infers the ambush layout from its position and orientation. The target is invisible. What it is attached to is not.